Design of Single Ended Sense Amplifier at Multiple Voltage Operation

Group no : 20

ECE-Memory Design and Test

INDRAPRASTHA INSTITUTE *of*
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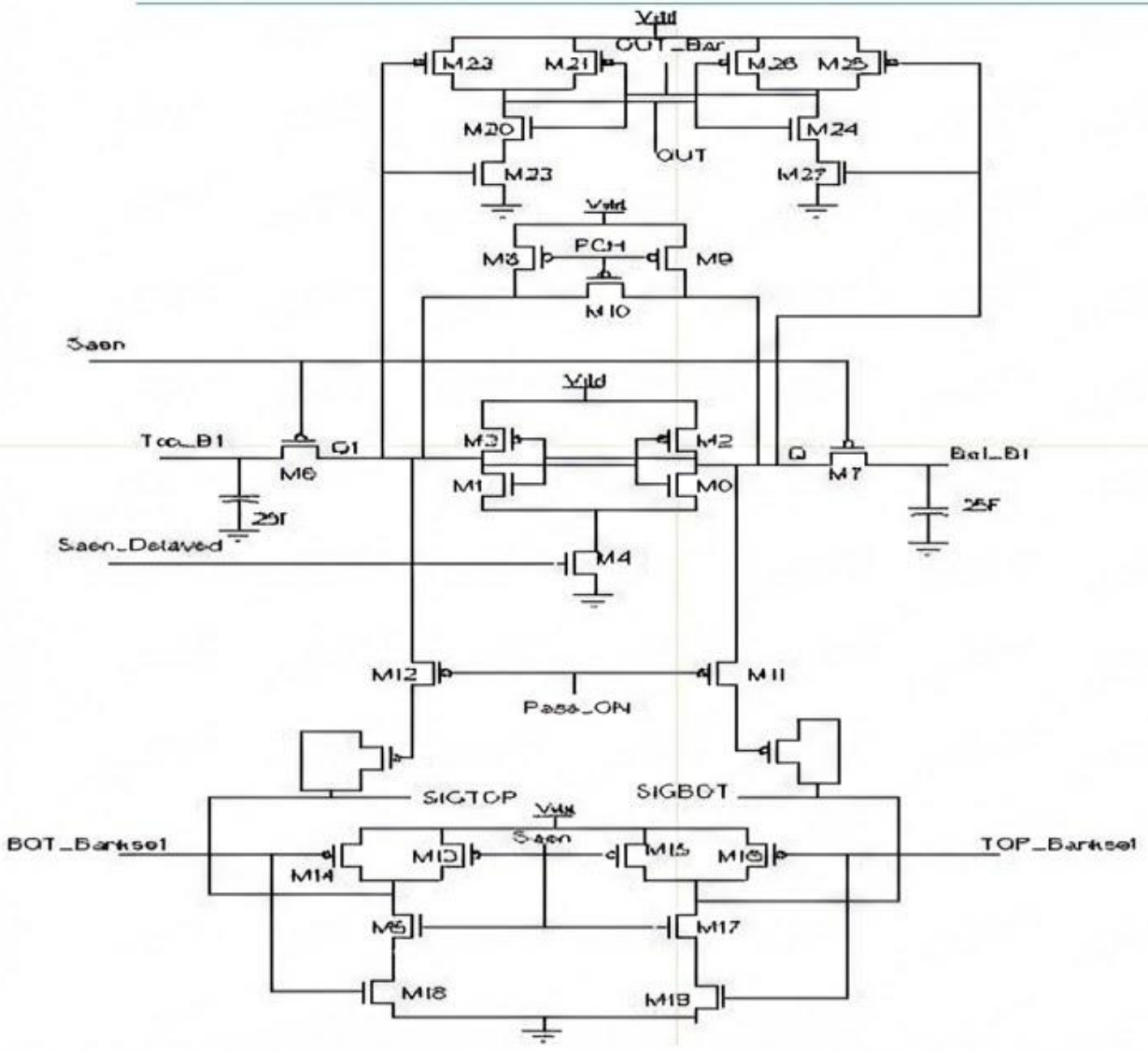
Problem Statement

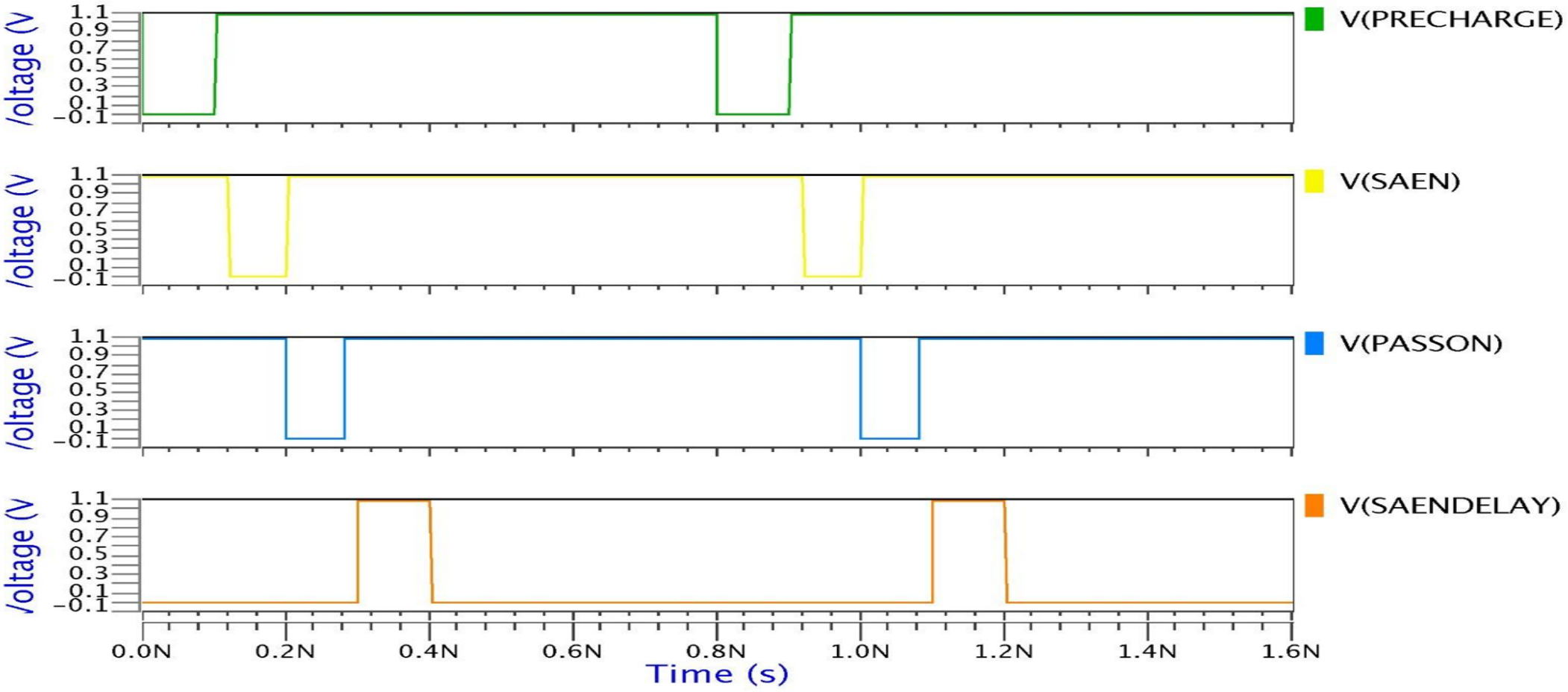
Challenge & Goal	Key Inference
1. Offset Voltage (3σ) within limit @ w_c, 1.08 V, 1.2 V, and 0.9 V	Post-layout input-referred offset must remain within 3σ limits under worst PVT conditions across all specified voltages.
2. Operation at 1.2 V, 1.08 V and 0.9V	Circuit should function reliably at both nominal and reduced supply voltages.
3. Pre and Post Layout Verification	Ensure performance metrics are met and compare results before and after layout implementation.
4. Response Time (200ps) @ Output Latch, 0.9 V	Circuit must achieve fast switching at the output to meet tight timing margins at the target voltage.
5. Normal and Common Centroid Layout Implementation	Develop and compare both layout strategies for optimal performance and matching.

Schematic and Sizing

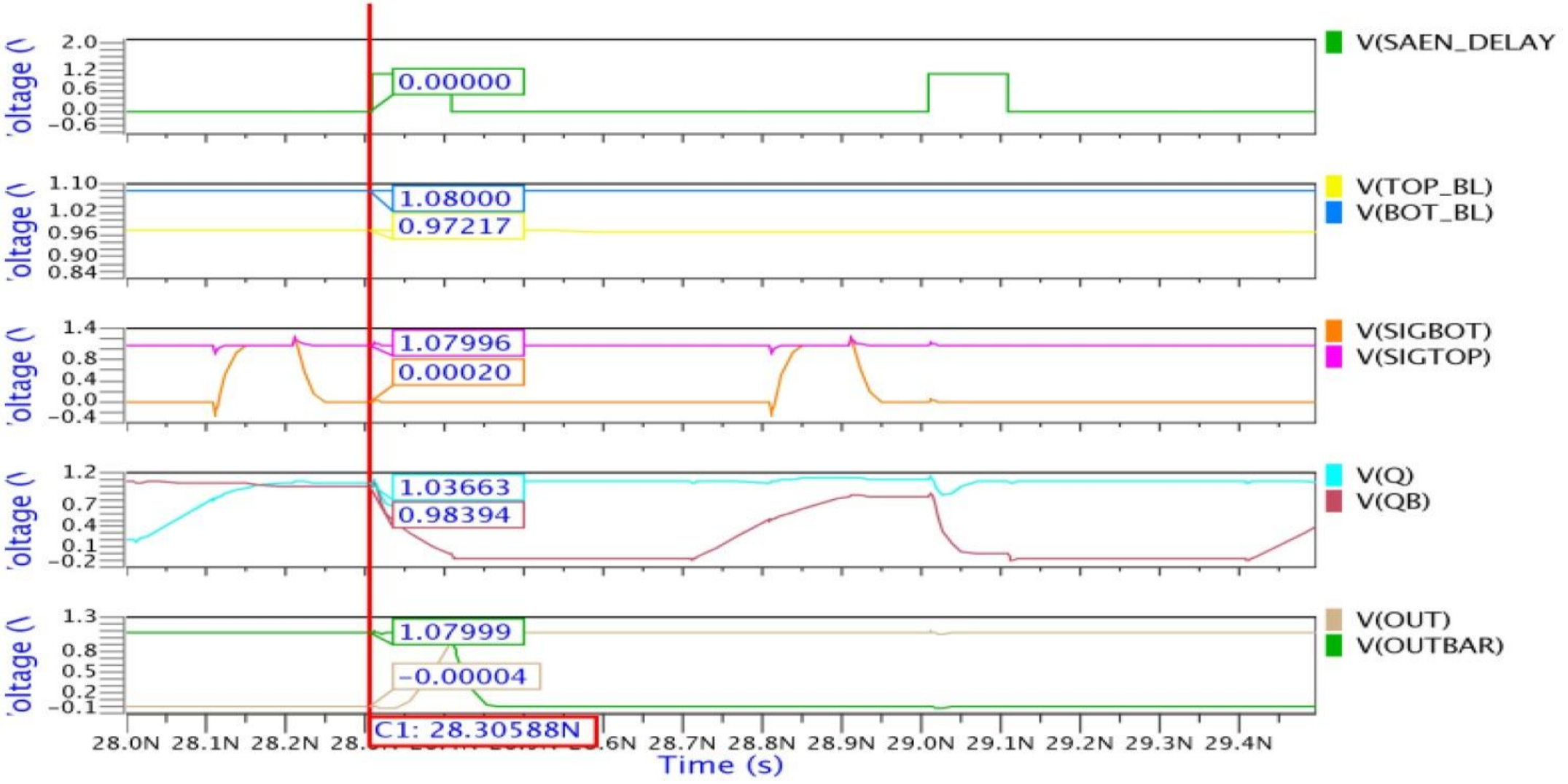


Transistor	Function Block	Type	W (μm)	L (μm)
M22, M21, M26, M25	Output latch	PMOS	0.27	0.06
M20, M23, M24, M27	Output latch	NMOS	0.27	0.06
M3, M2	Sense amplifier latch	PMOS	1.6	0.06
M1, M0	Sense amplifier latch	NMOS	2.0	0.06
M4	Tail transistor (footer)	NMOS	2.1	0.06
M8, M9	Precharge	PMOS	0.135	0.06
M10	Equalization (precharge)	PMOS	0.135	0.06
M6, M7	Pass gates (bitline access)	NMOS	0.27	0.06
M12, M11	PassON / charge sharing	NMOS	0.27	0.06
M13, M14, M15, M16	NAND logic	PMOS	0.27	0.06
M5, M17, M18, M19	NAND logic	NMOS	0.27	0.06

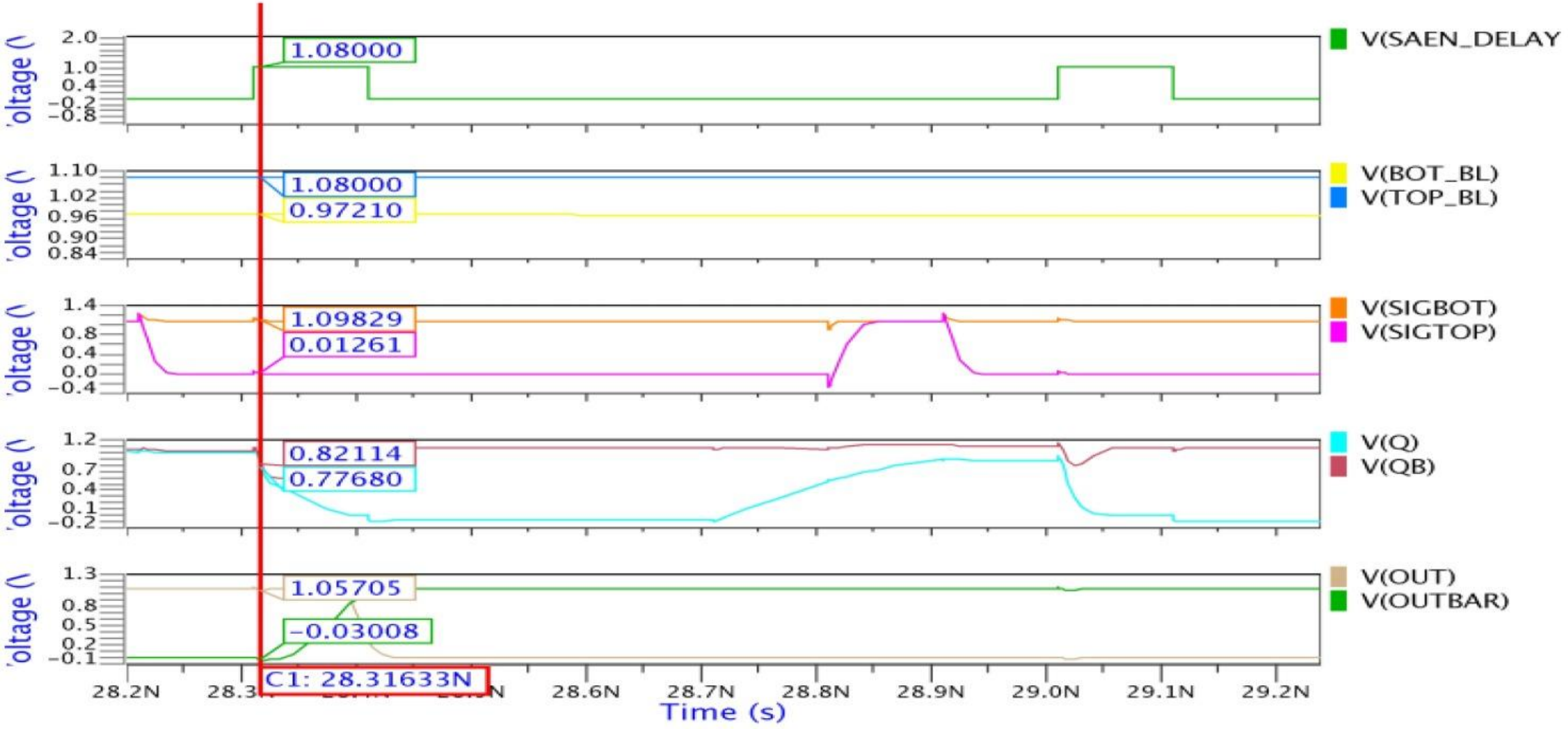
Verification Plan



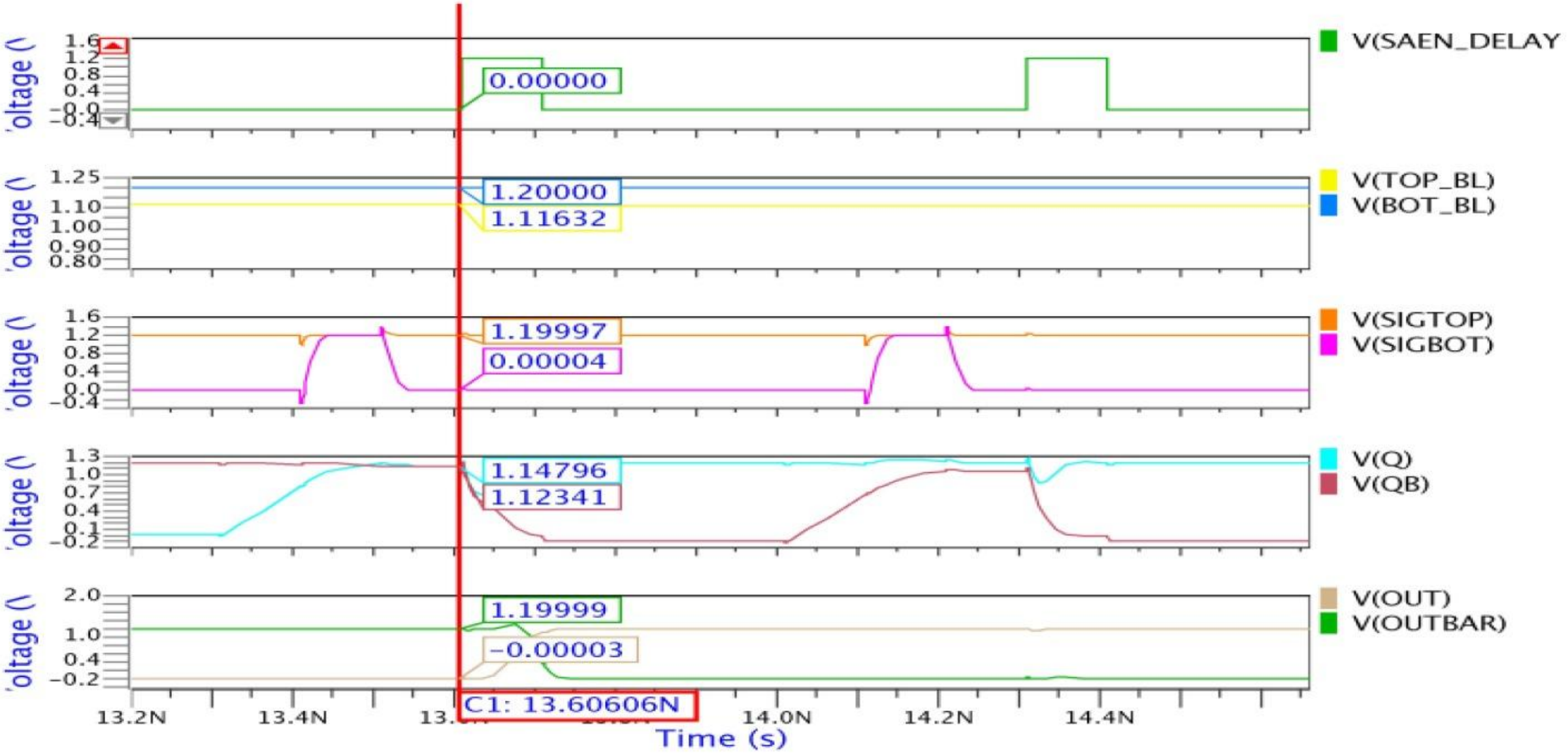
SA Functionality : Reading 0 (Focused view) 1.08v



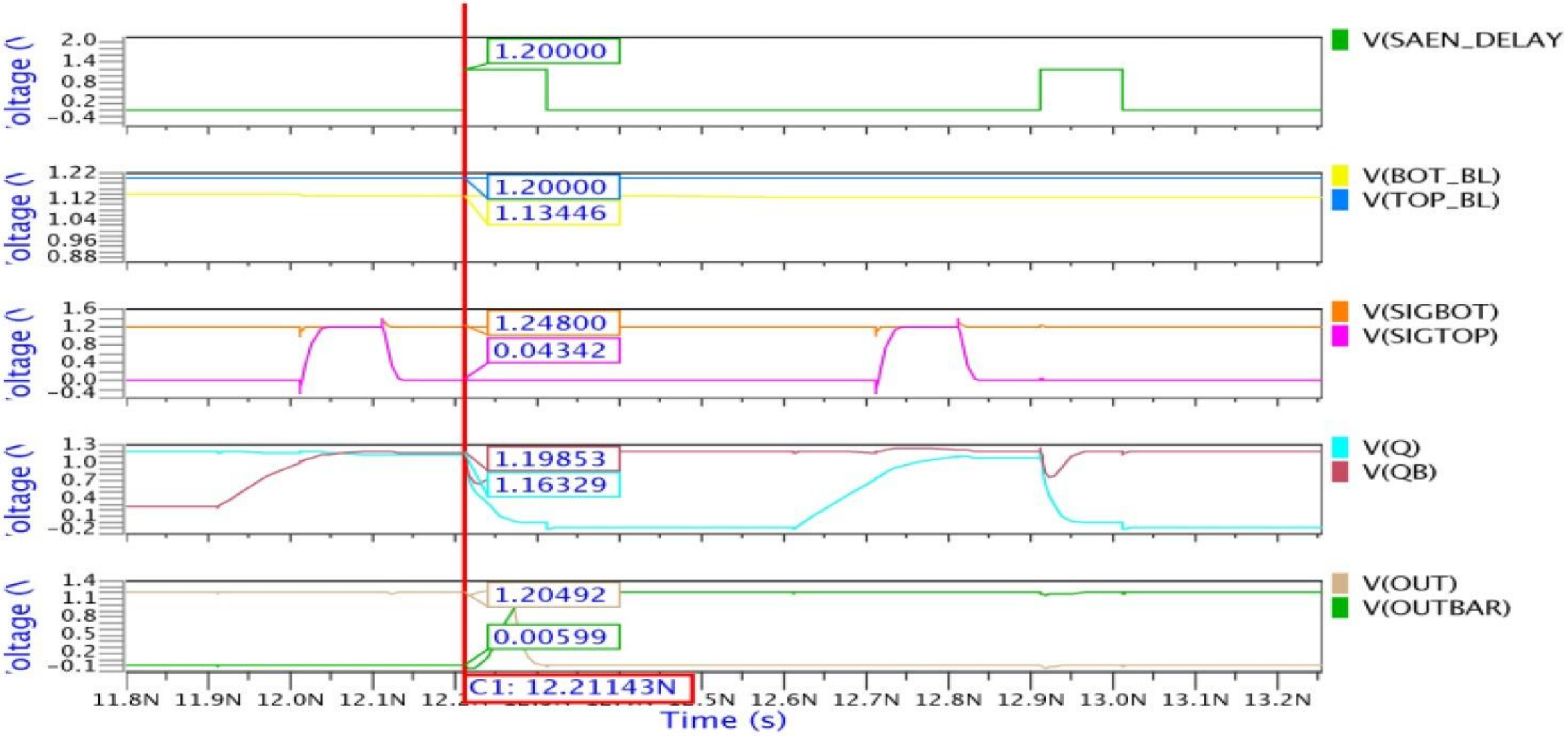
SA Functionality : Reading 1 (Focused view) 1.08v



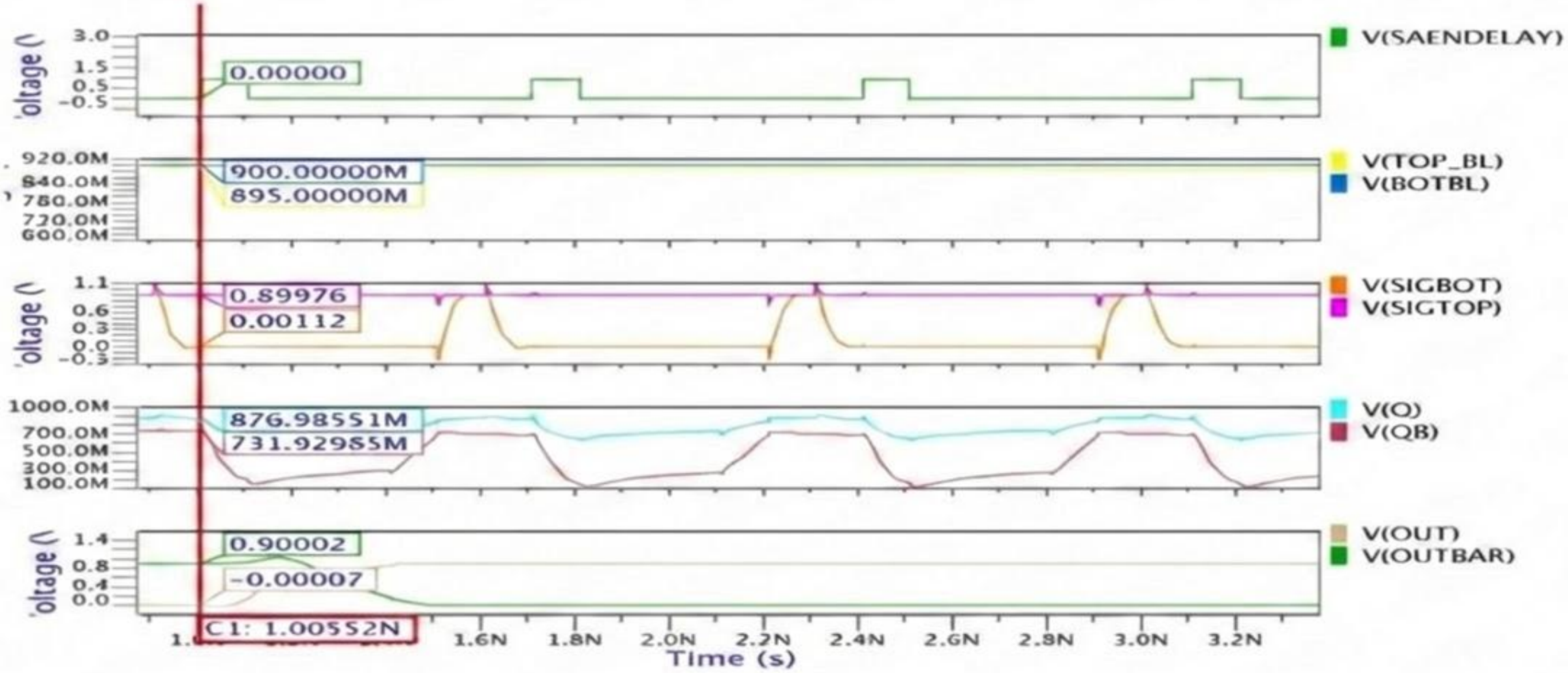
SA Functionality : Reading 0 (Focused view) 1.2v



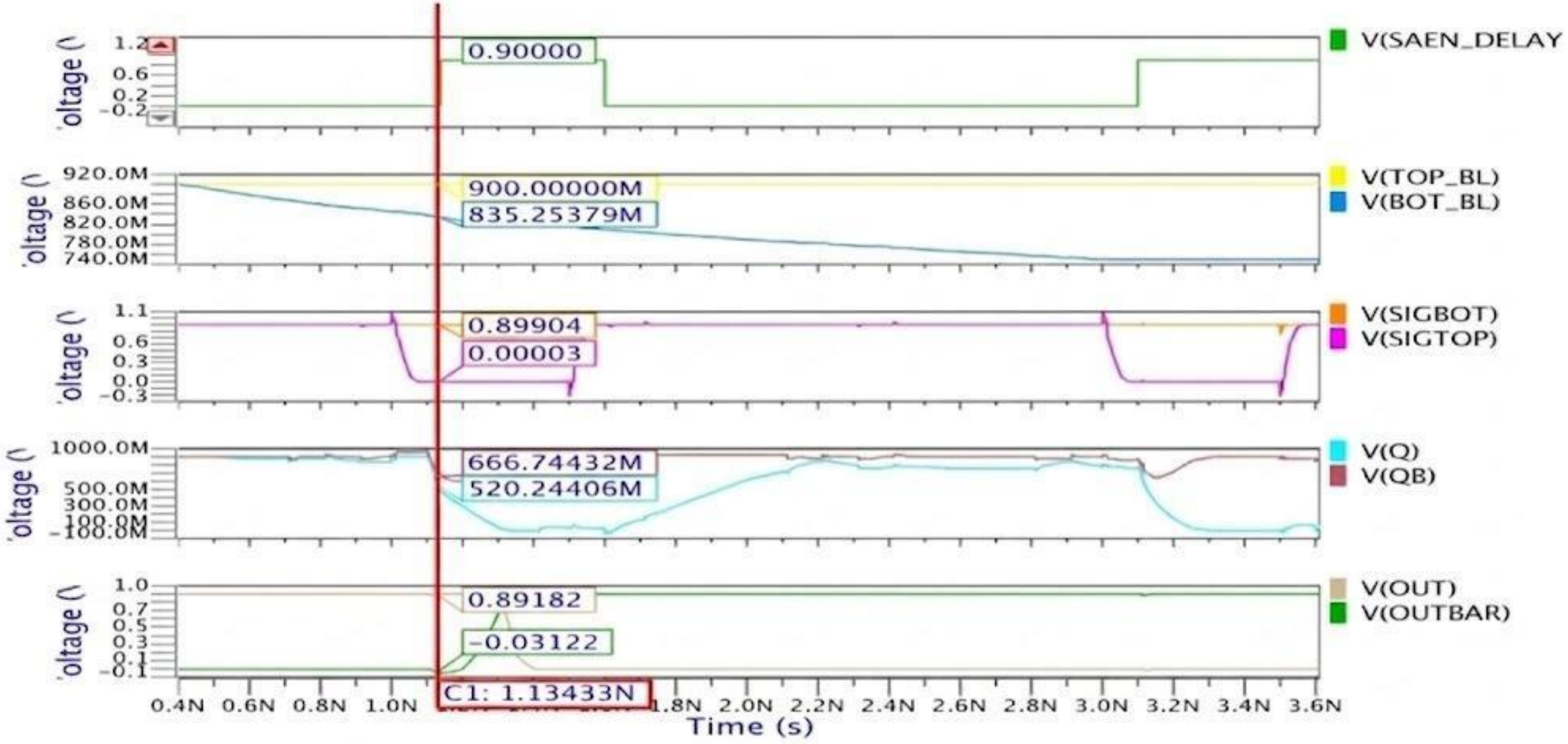
SA Functionality : Reading 1 (Focused view) 1.2v



SA Functionality : Reading 0 (Focused view) 0.9v



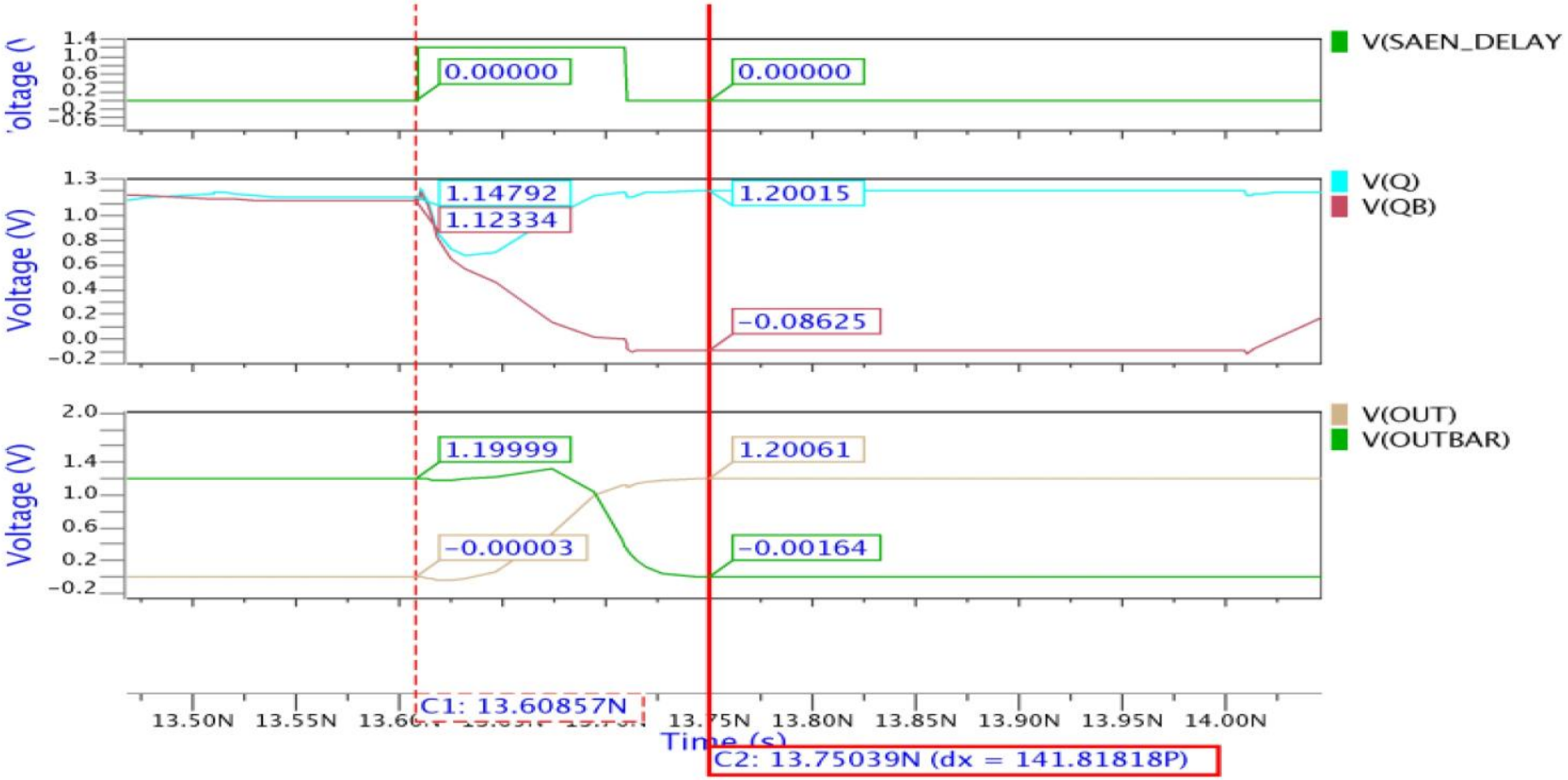
SA Functionality : Reading 1 (Focused view) 0.9v



SA Performance Summary Table

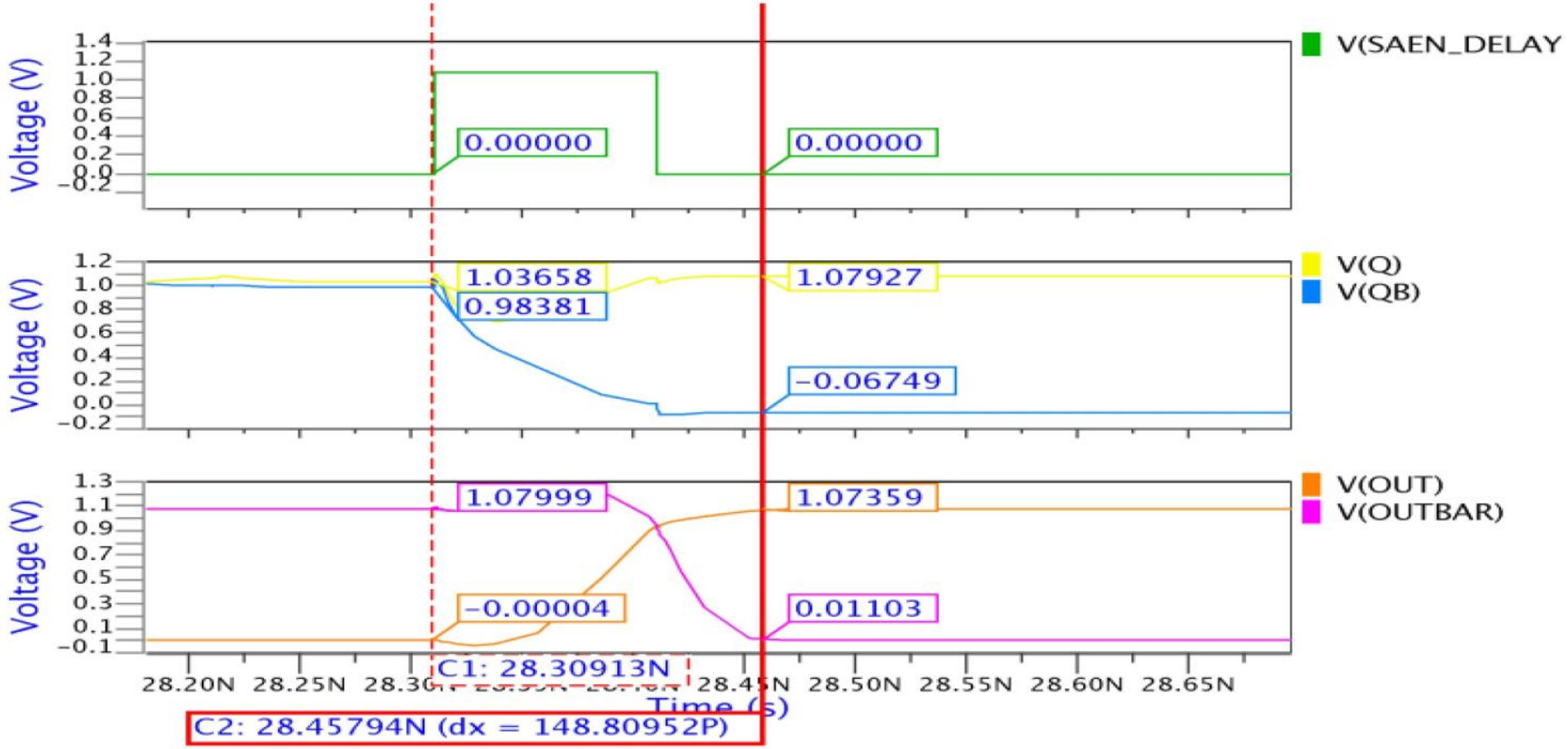
VDD (V)	Read Operation	Offset (mV)
0.9	Read 0	145.05
0.9	Read 1	139.16
1.08	Read 0	52.69
1.08	Read 1	44.34
1.2	Read 0	24.55
1.2	Read 1	35.24

Response Time 1.2V



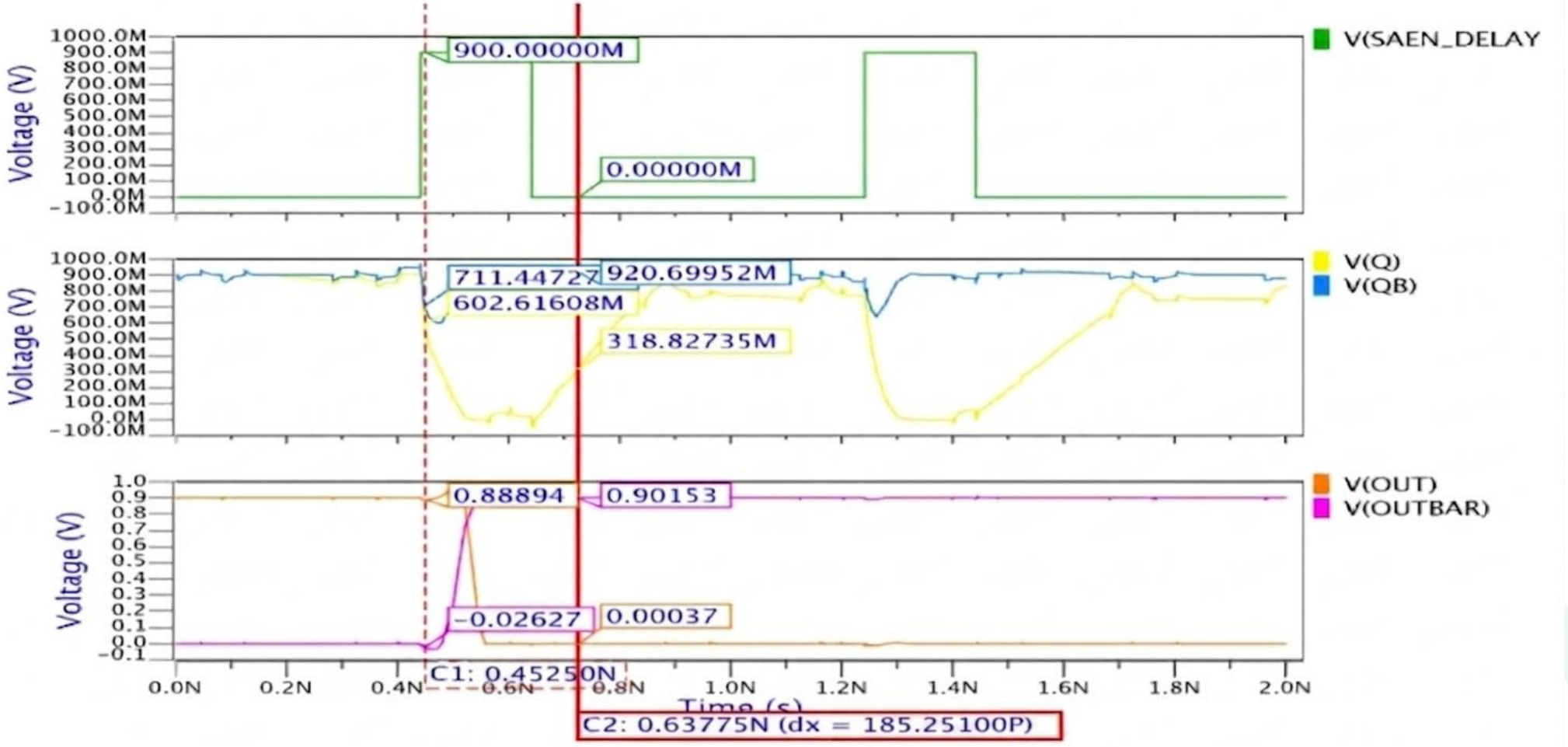
SAEN to Q latch delay for Read 0 =141.818 Ps for 1.2v

Response Time 1.08v



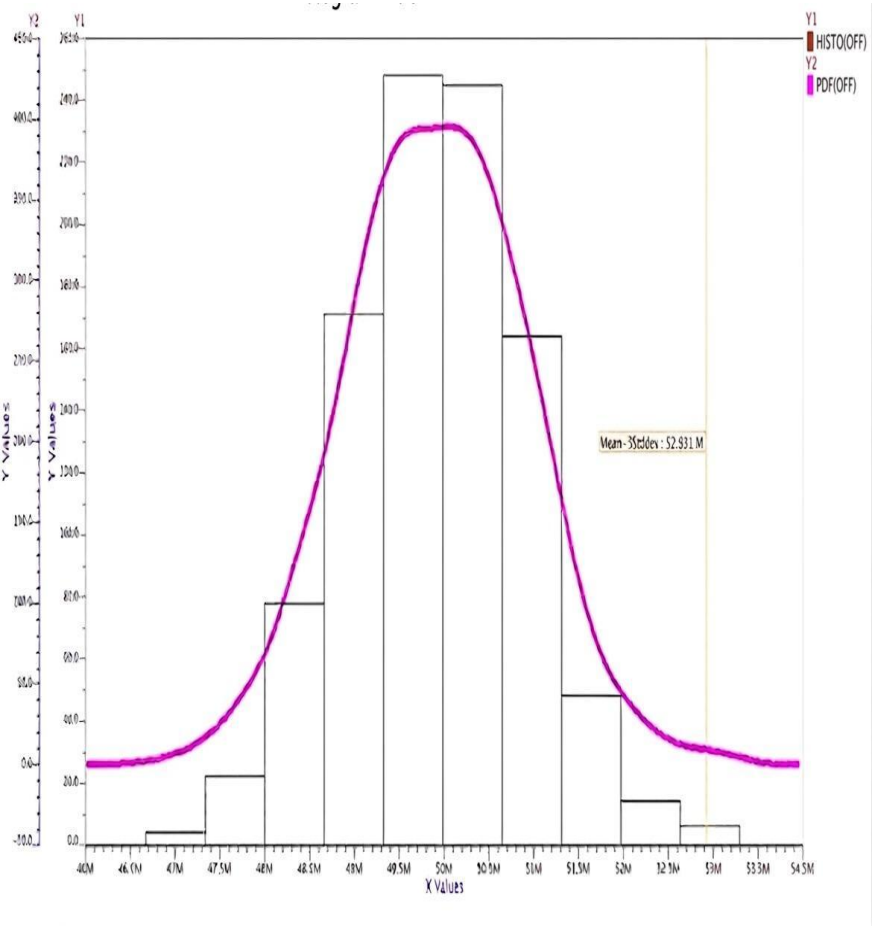
SAEN to Q latch delay for Read 0 =148.809 Ps for 1.08v

Response Time 0.9V

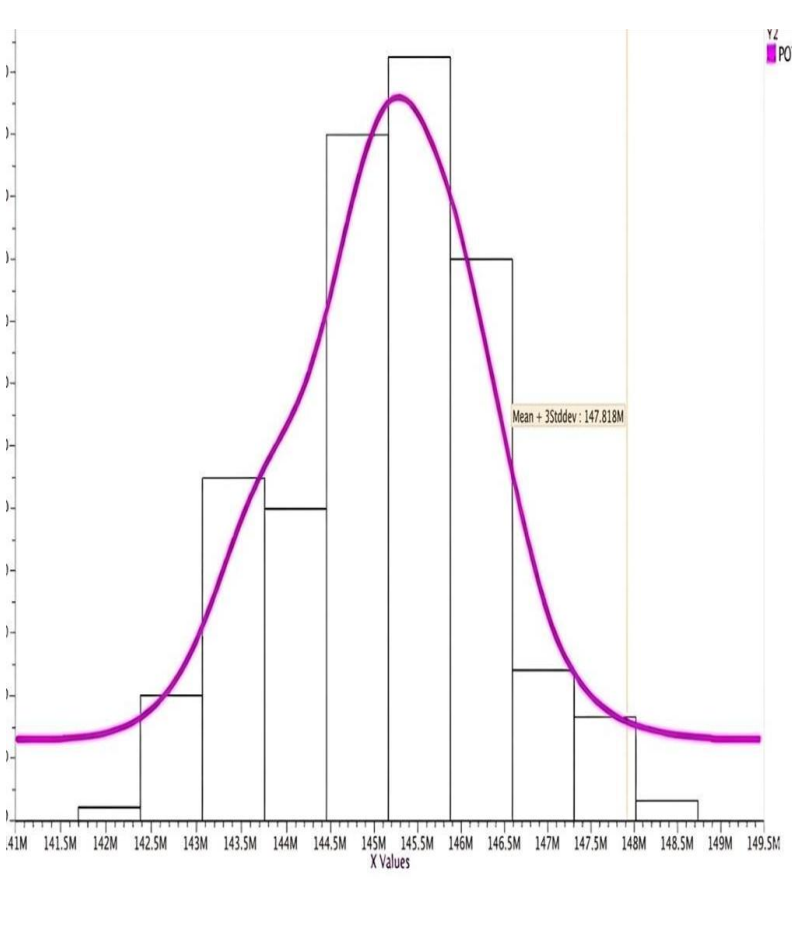


SAEN to Q latch delay for Read 0 =185.251 Ps for 0.9 v

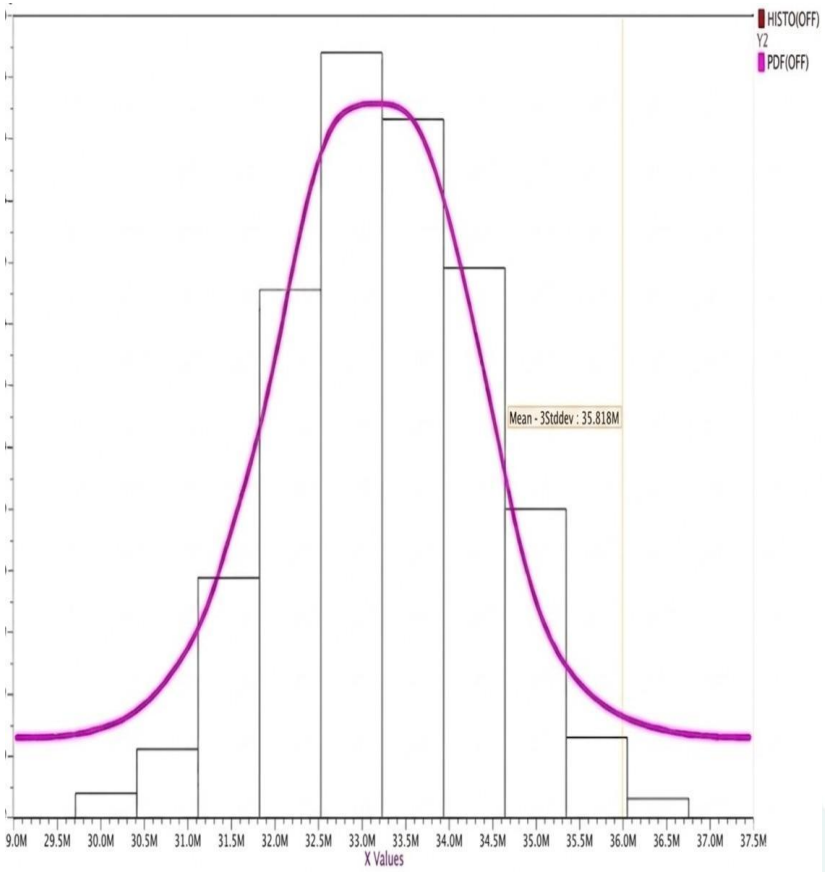
Offset Characterisation 3 Sigma@ WC (Monte Carlo)



1.08 V



0.9V



1.2V

Normal layout

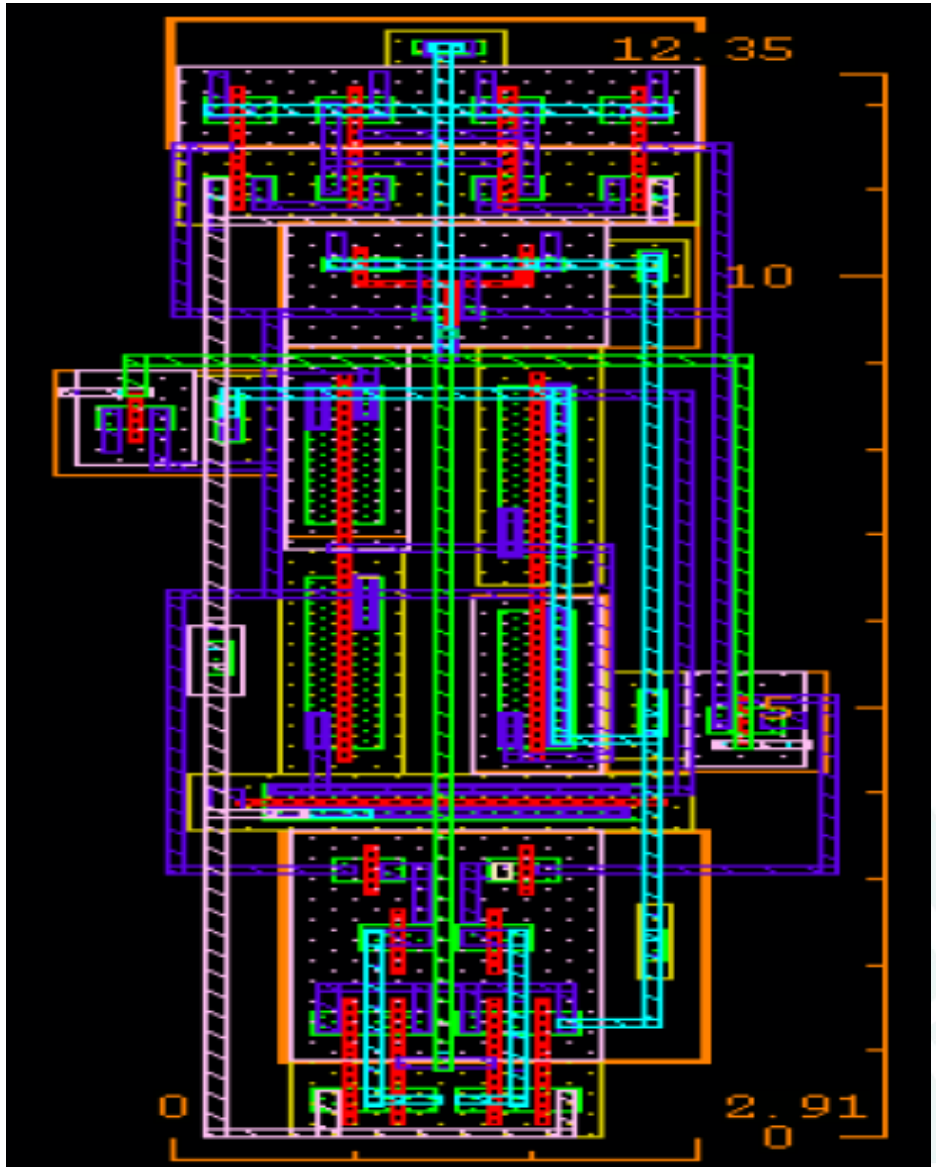
Layout: (Area = 35.9385 µm²)

Advantages:

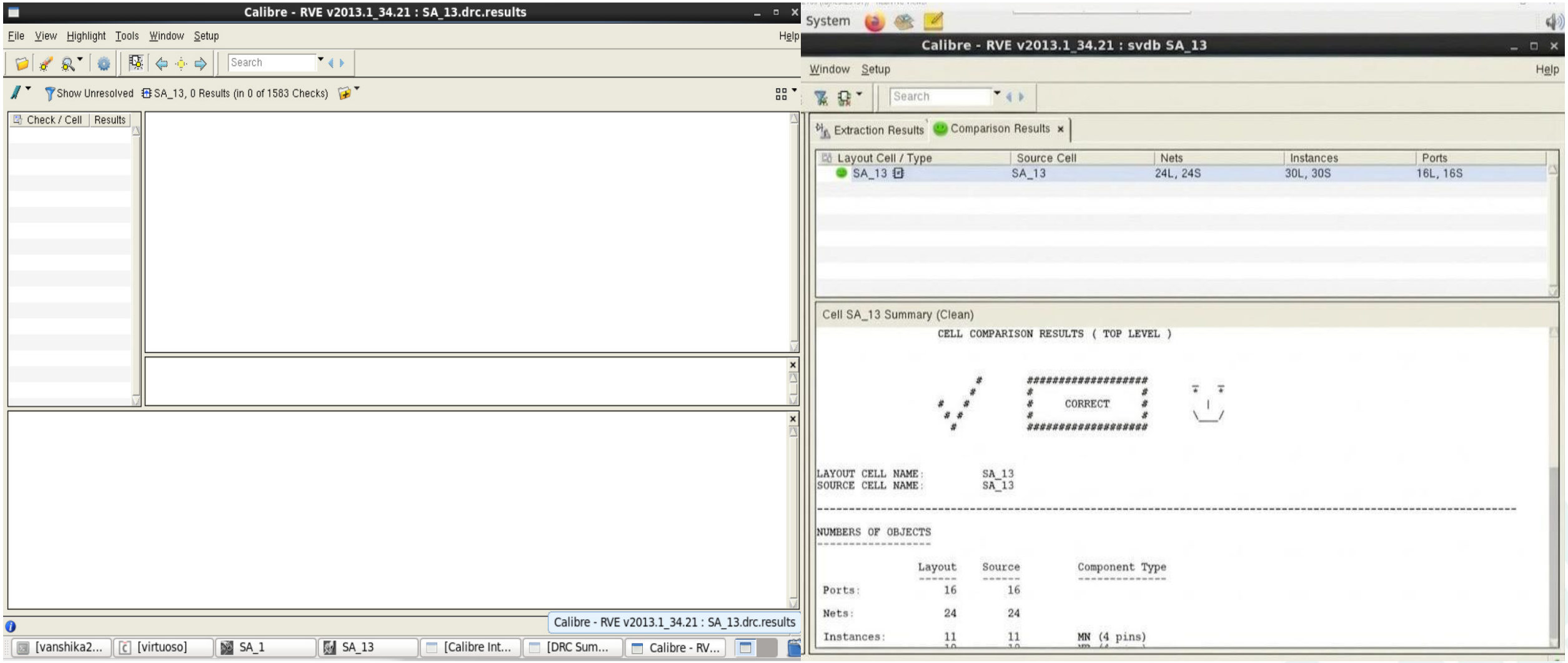
- **Compact coupled latch structure** → minimizes silicon usage, making it suitable for designs where symmetry is not priority
- **Simple layout** → low design complexity and faster implementation

Disadvantages:

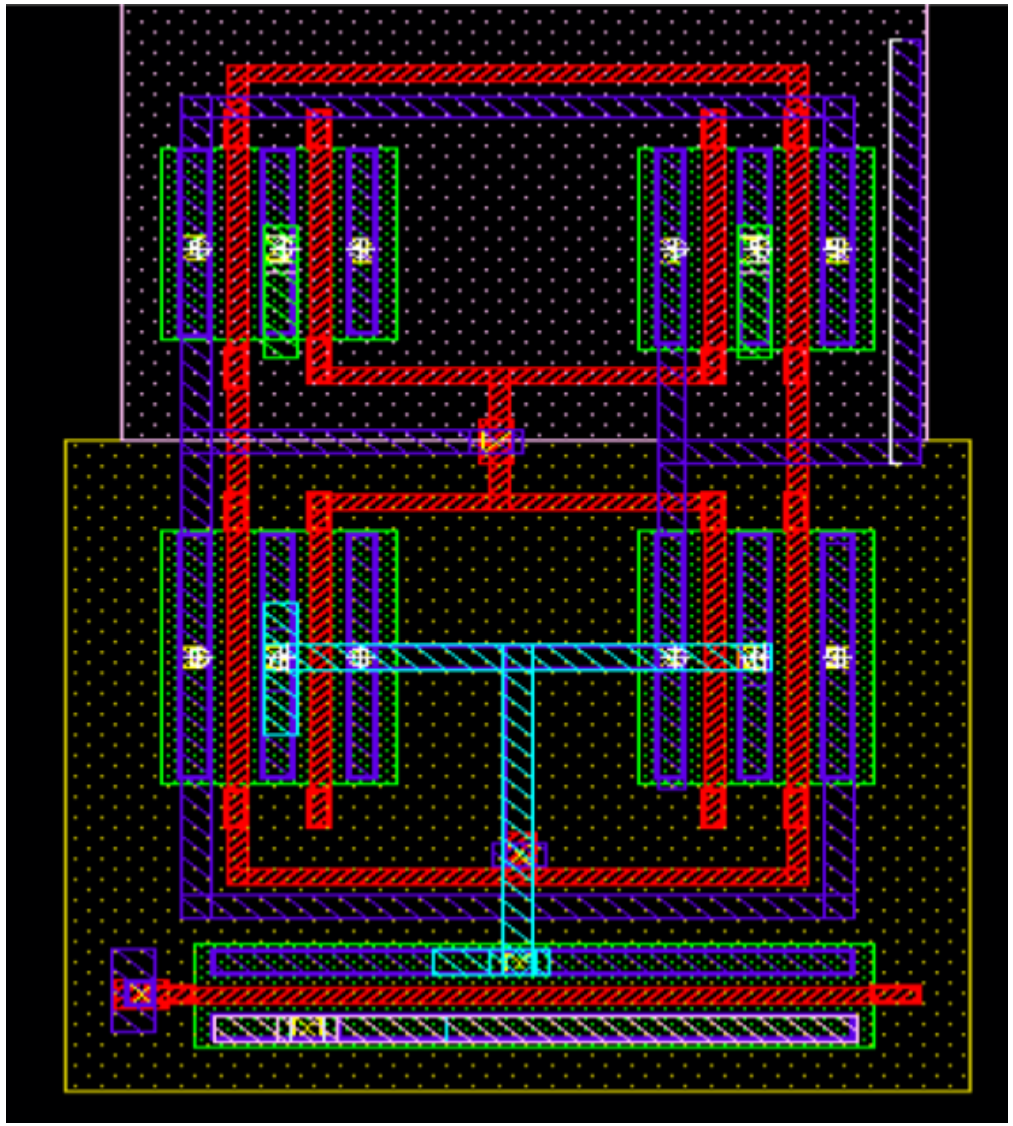
- **Mismatch** → higher offset due to poor symmetry
- **Parasitics** → routing imbalance affects delay and accuracy



DRC and LVS Clean



Common Centroid Layout (Cross Coupled Latch)



Common Centroid Layout for 1.08 V

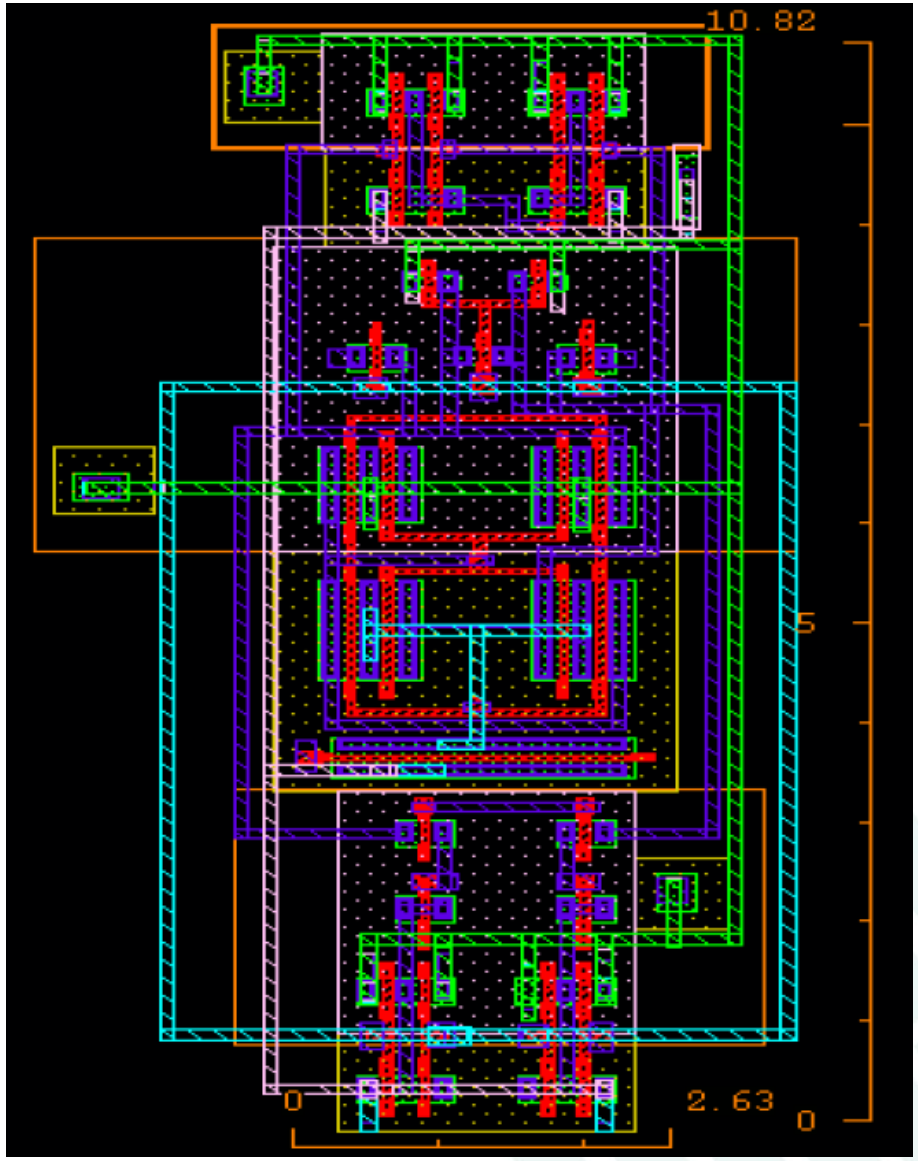
Layout: Area = 28.4566 µm²

Advantages:

- **Excellent matching** → centroid-based placement reduces mismatch and offset
- **Balanced parasitics** → symmetric routing improves speed and signal accuracy
- **Robust to PVT** → uniform device distribution ensures stable performance

Disadvantages:

- **Layout complexity** → requires careful planning and interdigitation



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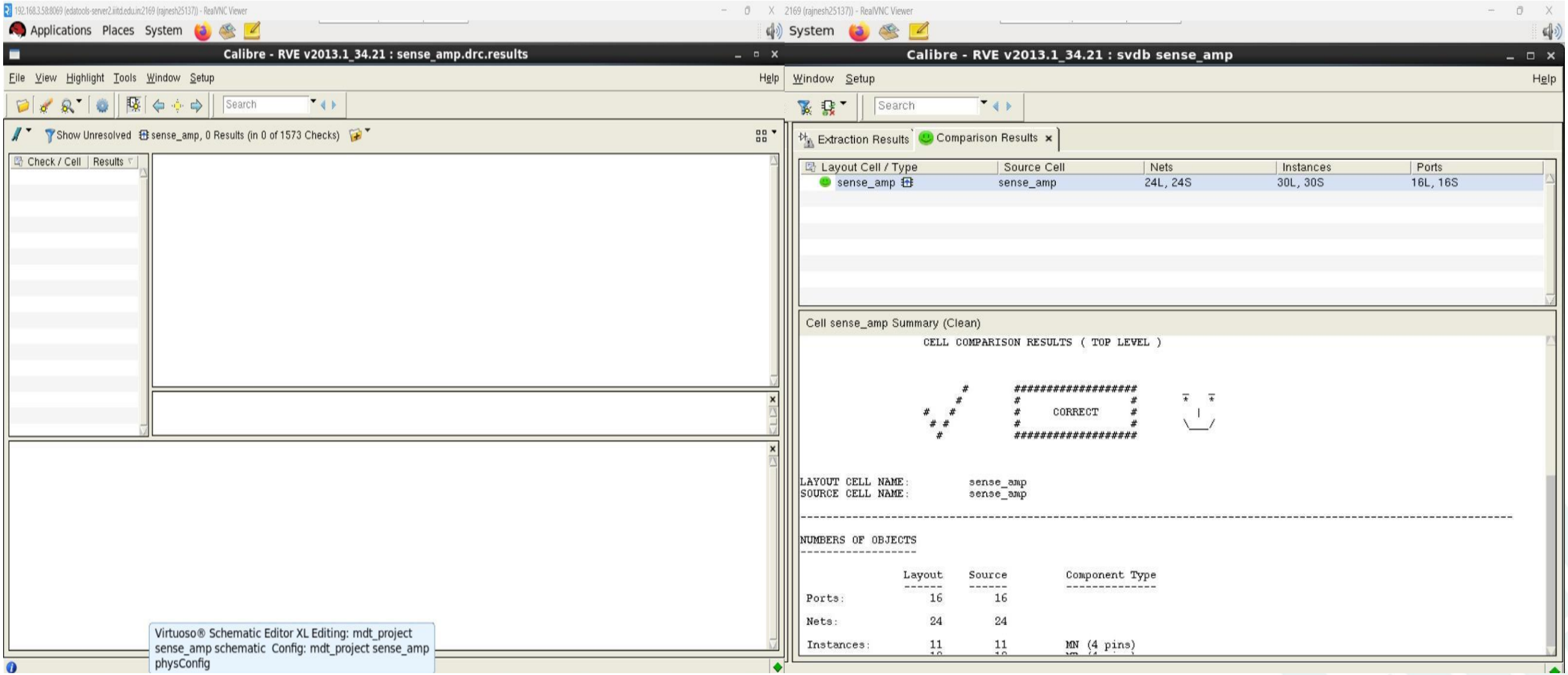


Comparison of Pre-Layout with Post-Layout Simulations 1.08V

PVT Condition	Pre-Layout (mV)	Post-Layout (mV)
SS, -40°C	49.95	55.80
SF, 125°C	12.20	16.50
SF, -40°C	4.80	7.20
FF, -40°C	11.50	14.20
FF, 125°C	2.90	4.10
FS, -40°C	28.60	32.80

Comparison of Normal and Common Centroid Layout

PVT Condition (Corner, Temp)	Post (Common Centroid) (mV)	Post (Normal Layout) (mV)
SS, -40°C	55.80	83.28
SF, 125°C	16.50	24.63
SF, -40°C	7.20	10.75
FF, -40°C	14.20	21.19
FF, 125°C	4.10	6.12
FS, -40°C	32.80	48.96

Common Centroid Layout for 0.9 V

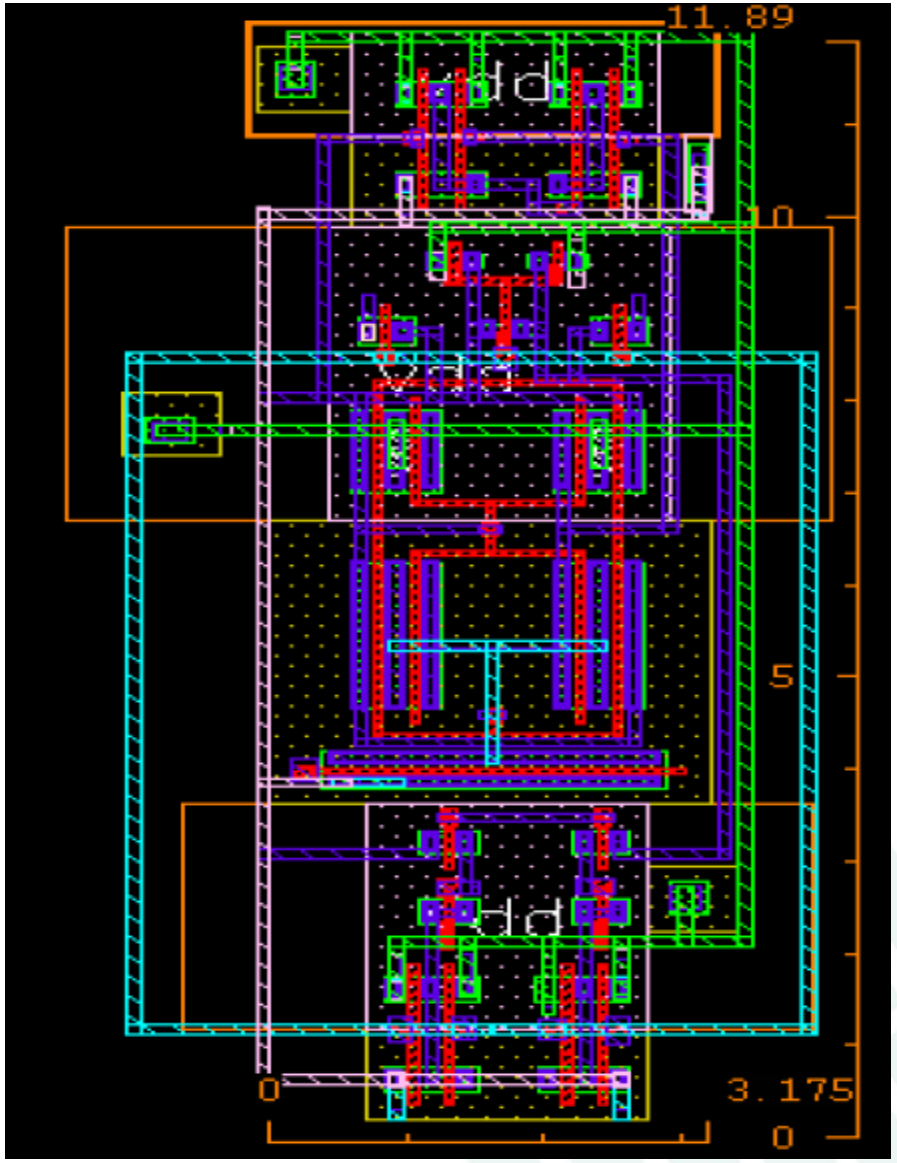
Layout: Area = 37.75 µm²

Advantages:

- **Excellent matching** → centroid-based placement reduces mismatch and offset
- **Balanced parasitics** → symmetric routing improves speed and signal accuracy
- **Robust to PVT** → uniform device distribution ensures stable performance

Disadvantages:

- **Layout complexity** → requires careful planning and interdigitation



Comparison of Pre-Layout with Post-Layout Simulations 0.9V

PVT Condition	Pre-Layout (mV)	Post-Layout (mV)
SS, -40°C	145.05	162.80
SF, 125°C	118.60	134.15
SF, -40°C	139.25	156.40
FF, -40°C	112.40	128.90
FF, 125°C	74.15	91.50
FS, -40°C	135.80	152.35

Conclusion

- Successfully designed and simulated a Single-Ended Sense Amplifier (SESA) across 1.2V, 1.08V, and 0.9V supply voltages.
- The common-centroid layout achieves superior device matching and significantly minimizes offset relative to conventional layouts, offering enhanced precision despite increased design complexity and layout overhead
- Response time and offset performance were thoroughly characterized through PVT corner analysis.
- Post-layout simulations confirmed the design's functionality and performance alignment with pre-layout expectations.
- The project demonstrates the critical trade-off between area, voltage, and complexity in analog design.

Work Distribution

Rajnesh Kumar

Pre-layout and post layout simulation

3 sigma, response time

Common-centroid layout

Normal layout

Khelan Vaishnav

X-circuit, Schematic

Circuit analysis

Verification plan

Post simulation

Rohit Bhardwaj

Sizing, Schematic

cir file, normal layout

Presentation

Vanshika Sharma

Sizing

Verification plan

Presentation